

Lesson 1 — Logistics & First Run

Matches LEGO Teacher Portal lesson: "Castle Quest".

Objective (I can...): I can pair the hub over Bluetooth, find my way around the Coding Canvas, and run a 2-block program. **CS concept:** What a program is (an ordered list of instructions a machine follows) · **Builds on:** No prior lesson — course launch.

AI Connection (2 min): Your phone unlocking with your face is a program plus AI making a choice. *First-principles question:* "If a computer had to 'recognize a cat,' what small steps might it break that into?" (Just get ideas — we'll build toward answering it all year.)

Materials: LEGO CS & AI kit (1 per pair), a device per pair (Coding Canvas), a printed "kit parts" picture card per pair. **No full build today** — use the kit's minimal/pre-built starter model (hub + one motor already attached, or the quick-start model). The unit's real model is built next lesson.

60-minute flow - Warm-up (8): "Robot chef" — a student gives the coach spoken steps to make a pretend sandwich; the coach follows them literally, mistakes and all. Point: machines do exactly what you say. - **Concept + demo (10):** Turn on the hub and show pairing it over Bluetooth. Tour the Canvas: block palette, workspace, Run and Stop. Drag a start block + one motor block; press Run so a real motor moves. - **Build/Code (30):** Each pair (1) pairs their hub over Bluetooth, (2) takes the guided tour — find the palette, Run, Stop, (3) places 2 blocks (start + spin a motor) on the pre-built model and runs it. *Swap Driver/Navigator at 15 min.* Coach circulates to confirm every hub is paired. - **Challenge (7):** Add a third block so the motor stops after a moment. Change the order and predict before running. - **Share + pack-up (5):** Each pair runs their program and names one kit part. Cables coiled, hub off, parts in tray.

Pair roles: Driver pairs the hub and drags blocks; Navigator holds the parts card and finds each part. Swap at 15 min. (*Trio: add a Tester who presses Run and reports what happened.*)

Differentiation: *Support* — give a labeled palette card so kids can find blocks by picture. *Extension / fast-finisher* — add a sound block when the motor stops. *Reteach* — re-pair the hub together slowly; have the child name each button before pressing.

Exit check: Pair has a paired hub, ran a 2-block program that moves a motor, and can point to the hub, a motor, and the Run button. **At-home (optional):** Write 5 exact steps for brushing teeth as if for a robot. ~10 min.